

Practical No. 8

A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user ie. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames 3).

```
/c/Users/acom/Desktop/OS_CD24037
GNU nano 8.7
#include <stdio.h>

int main() {
    int pages[50], frames[10];
    int n, framesCount = 3;
    int i, j, k, pos, faults = 0, hits = 0;
    int flag1, flag2;

    printf("Enter number of pages: ");
    scanf("%d", &n);

    printf("Enter page reference string:\n");
    for(i = 0; i < n; i++) {
        scanf("%d", &pages[i]);
    }

    for(i = 0; i < framesCount; i++)
        frames[i] = -1;

    for(i = 0; i < n; i++) {
        flag1 = flag2 = 0;

        for(j = 0; j < framesCount; j++) {
            if(frames[j] == pages[i]) {
                hits++;
                flag1 = flag2 = 1;
                break;
            }
        }

        if(flag1 == 0) {
            for(j = 0; j < framesCount; j++) {
                if(frames[j] == -1) {
                    faults++;
                    frames[j] = pages[i];
                    flag2 = 1;
                    break;
                }
            }
        }

        if(flag2 == 0) {
            int least[10];
            for(j = 0; j < framesCount; j++) {
                least[j] = -1;
                for(k = i - 1; k >= 0; k--) {
                    if(frames[j] == pages[k]) {
                        least[j] = k;
                        break;
                    }
                }
            }
        }
    }

    printf("Percentage Page Hit: %.2f\n", (float)hits/n*100);
    printf("Percentage Page Fault: %.2f\n", (float)faults/n*100);
}
```

AG Help AQ Write Out AF Where Is AK Cut
AX Exit AR Read File AN Replace AU Past

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 /c/Users/acom/Desktop/OS_CD24037

GNU nano 8.7

```
        flag1 = flag2 = 1;
        break;
    }
}

if(flag1 == 0) {
    for(j = 0; j < framesCount; j++) {
        if(frames[j] == -1) {
            faults++;
            frames[j] = pages[i];
            flag2 = 1;
            break;
        }
    }
}

if(flag2 == 0) {
    int least[10];
    for(j = 0; j < framesCount; j++) {
        least[j] = -1;
        for(k = i - 1; k >= 0; k--) {
            if(frames[j] == pages[k]) {
                least[j] = k;
                break;
            }
        }
    }

    pos = 0;
    for(j = 1; j < framesCount; j++)
        if(least[j] < least[pos])
            pos = j;

    frames[pos] = pages[i];
    faults++;
}

printf("Frame status: ");
for(j = 0; j < framesCount; j++)
    printf("%d ", frames[j]);
printf("\n");
}

float hitPercent = ((float)hits / n) * 100;
float faultPercent = ((float)faults / n) * 100;

printf("\nTotal Page Hits = %d\n", hits);
printf("Total Page Faults = %d\n", faults);
printf("Hit Percentage = %.2f%%\n", hitPercent);
printf("Fault Percentage = %.2f%%\n", faultPercent);

return 0;
}
```

 Help
 Exit

 Write Out
 Read File

 Where Is
 Replace

 Cut
 Paste



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 /c/Users/acom/Desktop/OS_CD24037

```
acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037
$ nano lru.c
```

```
acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037
$ gcc lru.c -o lru
```

```
acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037
$ ./lru
```

Enter number of pages: 12

Enter page reference string:

1 2 3 4 1 2 5 1 2 3 4 5

Frame status: 1 -1 -1

Frame status: 1 2 -1

Frame status: 1 2 3

Frame status: 4 2 3

Frame status: 4 1 3

Frame status: 4 1 2

Frame status: 5 1 2

Frame status: 5 1 2

Frame status: 5 1 2

Frame status: 3 1 2

Frame status: 3 4 2

Frame status: 3 4 5

Total Page Hits = 2

Total Page Faults = 10

Hit Percentage = 16.67%

Fault Percentage = 83.33%

```
acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037
$ |
```